

Whole-cell protein extraction and Western blotting

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1. Purpose

This protocol describes cell lysis for whole-cell protein extraction followed by western blotting using LI-COR Odyssey quantitative western blot. Frozen cell pellets described here are typically iPSC-derived cortical neurons, seeded into a 24 well plate format at 1×10^6 cells per well, and collected for downstream processing at DIV25.

2. Materials

iBlot 2 Gel Transfer Device (Thermo Fisher)
Odyssey CLx Infrared Imager (LI-COR)

3. Reagents

NuPage™ LDS sample buffer (Life Technologies, Cat. # NP0007)
cOmplete™ EDTA-free Protease Inhibitor Cocktail (Sigma Aldrich, Cat. # 11873580001)
PhosSTOP phosphatase inhibitor cocktail (Sigma Aldrich, Cat. # 4906845001)
Pierce™ BCA Protein Assay Kit (Thermo Fisher, Cat. # 23227)
NuPAGE 4-12% Bis-Tris protein gels (Life Technologies, Cat. # NP0322BOX)
NuPAGE MES SDS Running Buffer (Life Technologies, Cat. # NP000202)
4% Paraformaldehyde (Thermo Scientific, Cat. # 28906)
LI-COR Odyssey blocking buffer (PBS) (Fisher Scientific, Cat. # NC9877369)
Tween-20 (Fisher Scientific, Cat. # 50213612)
Primary antibody
IRDye secondary antibody (LI-COR): e.g., IRDye® 800CW (LI-COR Biosciences, Cat. # 925-32214), IRDye® 680RD (LI-COR Biosciences, Cat. # 925-68074)

4. Procedure

1. For cell lysis, thaw frozen cell pellets on wet ice
2. Resuspend cell pellets in 100 µL of 1X NuPage™ LDS sample buffer (diluted using dH₂O) containing protease and phosphatase inhibitors.
3. Sonicate samples twice for 15 sec with a tip-sonicator at 40% power, keeping sample on ice.
4. Centrifuge samples at maximal speed for 10 min.
5. Transfer the supernatant to a new Eppendorf tube. Discard the cell pellet.
6. To quantify protein concentration, 5 µL of each sample was used in Pierce™ BCA Protein Assay Kit according to manufacturer's guidelines.

- a. Note: Add 5 μ L of 1X LDS sample buffer containing protease and phosphatase inhibitors to bovine albumin serum (BSA) protein standards to account for LDS dye in absorbance measurement.
7. For western blots, set aside 30 μ g of protein per sample for SDS-PAGE using NuPAGE 4-12% Bis-Tris protein gels in NuPAGE MES SDS Running Buffer.
8. Conduct electrophoresis at 150 V for 55 min or until the protein ladder and the loading dye indicate a sufficient electrophoretic separation.
9. Conduct dry transfer from polyacrylamide gel to nitrocellulose membrane with the iBlot 2 Gel Transfer Device using preset P0 program (20V 1 min; 23V 4 min; 25V 2 min).
10. If detecting alpha-synuclein, fix membrane in 4% paraformaldehyde in dH₂O for 30 min at room temperature with orbital shaking. Membrane fixation improves alpha-synuclein detection (Lee et al. 2011 *PLoS ONE*).
11. Wash membrane three times for 5 min with PBS.
12. Block membranes in Li-COR Odyssey blocking buffer (PBS) for 1 h with orbital shaking.
13. Incubate membrane overnight in primary antibody solution, i.e. Odyssey blocking buffer (in PBS), 0.1% Tween-20 and the respective primary antibodies at the desired dilution, at 4°C with orbital shaking.
14. Wash membranes four times for 5 min in 0.05% Tween-20/PBS.
15. Incubate membranes with the secondary antibody solution, i.e. Odyssey blocking buffer (TBS), 0.1% Tween-20 and secondary antibody at 1:10,000 dilution, at room temperature with orbital shaking and protected from light.
16. Wash membranes four times for 5 min in 0.05% Tween-20/PBS.
17. Scan the blot using the Odyssey CLx Infrared Imager.
18. Use ImageStudio software for quantification of bands.

5. References

Lee, B.R., and Kamitani, T. (2011). Improved Immunodetection of Endogenous α -Synuclein. *PLoS ONE* 6, e23939. <https://doi.org/10.1371/journal.pone.0023939>.